



Habib University
shaping futures

Introduction to Probability and Statistics

EE/CE 354/361 L1

Fall Semester 2024

The house always wins

No	Item	Details
1	Course LMS (Canvas) URL – L1 section & L2	L1 - https://hulms.instructure.com/courses/3974
2	Instructional mode (details in section III)	In-person instruction with – 2 weekly sessions of 75 minutes each.
3	Course pre- requisites	(i) MATH 102 - Calculus II (ii) Python or some other programming language. Rudimentary linear algebra
4	Hardware pre-requisites	preferable - Laptop
5	Software pre-requisites	Canvas, MS teams, Acrobat Reader, MS office, Web browser.
6	Content Area	This course meets program requirements for EE, CE & CS Majors & Quantitative Reasoning (QR) Form of Thought for DSSE Majors
7	Campus Safety policy	Please read the campus safety policy

Instructor Information

Instructor: Ahmed Ali Mustansir

Office Location: N412

Email: ahmedali.mustansir@sse.habib.edu.pk

Office Hours: Thursday 1:30 PM to 3:30 PM

Course Description

In the present world, we encounter situations where we have to make decisions on the basis of incomplete or imperfect information. The theory of probability helps provide a formal mechanism for understanding, quantifying, and dealing with uncertainty, which is ever present in our lives, pure science, or engineering applications. Simply, by uncertainty we mean the condition when outcomes or future are not completely determined or can be captured by a deterministic function; they depend on a number of factors and perhaps just on pure chance. A lot of our present day technologies will not be possible without an understanding of how to make decisions in presence of uncertainty. These technologies include all forms of wireless communication, servers, speech processing systems, network systems and so many more. Equipping yourself with tools to deal with uncertainty will help you with whatever you wish to pursue in life.

Course Aims

This course will cover the foundations of probability, random variables and statistics, with a plethora of examples from electrical engineering, computer engineering, computer science, and everyday life. The course content can broadly be divided into three categories: -

1. Fundamentals of probability,
2. Common probability models,
3. Inferences & statistics.

We'll be working towards the following 'course objectives' in this course: -

1. Gaining an appreciation for the limitation of human intuition when faced with finding optimal solution for a problem in the presence of uncertainty.
2. Recognizing the presence of uncertainty in a scenario and modeling it using formal notions of probability.
3. Identifying the right questions to ask when drawing inferences from a probability model and interpreting results correctly.

Course Learning Outcomes (CLOs)

Course Learning outcomes		
After the completion of the course, students should be able to:		
SLO/ CLO	Description	Learning- Domain

		Level
CLO 1	Transform a description of random experiment into formal setting of probability theory – construct sample space, identify and specify events.	Cog - 4
CLO 2	Apply the axioms of probability and compute probabilities by counting.	Cog - 3
CLO 3	Compute probabilities of events and moments from given distributions.	Cog - 3
CLO 4	Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables;	Cog - 3
CLO 5	Analyze a random experiment for presence of dependent situations and modeling and analyzing it using notions of conditional probability.	Cog - 4

Mode of Instruction

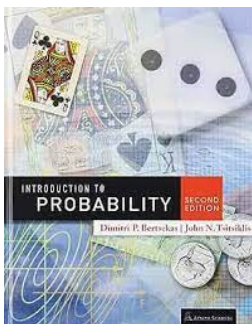
Nature of Instruction:

In person lecture twice a week

Engagement & Participation Rules

Course participation is an important aspect for this course – therefore, one key guide is to give 100% attention to the lectures. There will be multiple engagement opportunities during the course, for example, in the form of open questions. You are expected to participate in all of these activities.

Required Texts and Materials

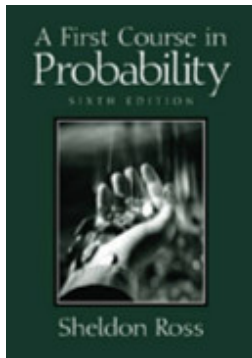


Introduction to Probability

Authors: Dimitri P. Bertsekas, John N. Tsitsiklis

Publication Date: 2002-01-01

Optional Materials



A First Course in Probability

Edition: 10

Authors: Sheldon M. Ross

Assessments

Assessments - Break Down of Marks (minor tweaking could be made)

No	Assessment type	Total	Weight (%)	Remarks
1	Short 15-20 minutes quiz	8	30	Weekly or fortnightly quizzes of 15 mins each.
2	Assignments	4	16	Two assignments on module No 1 (chapter No 1), one assignment from module 2 (chapter No 2) & one assignment from module 3 (Chapter No 3)
3	Programming assignments	1-2	10	Python based assignments to be done in pairs
4	Mid-Term Exam	1	20	An in-person exam will be conducted that would be comprehensive with the material taught till end of week No 7
5	End-term Exam	1	20	An in-person exam will be conducted that would be comprehensive.
6	Case study	1	4	To be done in groups. The mini project will use all skills you have to learnt in this course to solve a real world problem

Grading Scale

Letter Grade	GPA Points	Percentage
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Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Week-Wise Schedule (Tentative)

Week No	Topics to be covered	Out	In
1	Introduction; Sets, Sample Space, Probabilistic Models, Axioms,		
2	Discrete & Continuous Models; Conditional Probability;	Quiz 1	Quiz No 1
3	Total Probability; Bayes' rule	Assignment 1	
4	Independence; Counting	Quiz 2	Quiz 2
5	Discrete Random variables, PMF; Common discrete RVs	Quiz 3	Quiz 3 Assignment 1
6	Common discrete RVs; Functions of RVs;	Quiz 4 Assignment 2	Quiz 4
7	Mean, and Variance of known Discrete Random Variables; Joint PMF	Quiz 5	Quiz 5
8	Conditional PMF; Conditional Expectation; Independence of RVs	MT	Assignment 2 MT
9	Continuous RVs and PDF; Expectation; CDF		

		Assignment 3	
10	Normal RVs;	Quiz 6	Quiz 6 Assignment 3
11	Joint PDFs;	Assignment 4	
12	Conditional PDF's	Quiz 7	Quiz 7 Assignment 4
13	Conditional PDF's	Programming assignment	
14	Continuous Bayes' Rule	Quiz 8 Case study	Quiz 8 Programming assignment
15	Continuous Bayes' Rule		
16	Derived distributions;		Case study
17	End-Term Exam	ET	ET

Note – Due to dense assignments, the instructors will try their best to return the graded instruments within two weeks' window. In some cases, the grading might take an additional one week.

Final Exam Policy

An in-person final exam will be conducted that would be comprehensive (in terms of content).

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Late Submission Policy

Late submission of Assignments – Late submissions will be penalized at the rate of 0.5% per hour late upto 2 days of late submissions after which the score is an automatic 0.

- Required & Reference Books, Presentations and Support Material
 1. All course presentations, recorded sessions, lecture slides, required textbook, reference books, articles and all other support material will be managed within the CANVAS site.
 2. All assignments, quizzes and assessments with instructor solution would be managed through the CANVAS course site.

Attendance Policy

Attendance - 100% attendance is required. Because of any emergency, attendance required must be kept above 85% to continue with the enrolment. For any missed class an email from the student is expected – preferably prior to missing the class

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Engineering Knowledge: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
- PLO 2: Problem Analysis: An ability to identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

	Program Learning Outcomes (PLOs) mapped to MATH 310/EE 354 SLOs/CLOs
	SLOs/CLOs of the course are designed to cater following PLOs: PLO 1 : Engineering Knowledge, PLO 2 : Problem Analysis

		Distribution of CLO weights for each PLO			
PLOs	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1	25%	25%	25%	25%	
PLO 2					100%

	CLO#01	CLO#02	CLO#03	CLO#04	CLO#05
Assign 1					
Assign 2					
Assign 3					
Assign 4					
Quiz No 01					
Quiz No 02					
Quiz No 03					
Quiz No 04					
Quiz No 05					
Quiz No 06					
Quiz No 07					
Quiz No 08					
Mid-Term Exam – Q1					
Mid-Term Exam – Q2					
Mid-Term Exam – Q3					
End-Term Exam – Q1					
End-Term Exam –					

Q2					
End-Term Exam – Q3					
End-Term Exam – Q4					

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature

- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.